Spatial–Spectral Graph Contrastive Clustering With Hard Sample Mining for Hyperspectral Images

Renxiang Guan, Wenxuan Tu, Zihao Li, Hao Yu, *Student Member, IEEE*, Dayu Hu, Yuzeng Chen, Chang Tang, *Senior Member, IEEE*, Qiangqiang Yuan, *Member, IEEE*, and Xinwang Liu, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) clustering is a fundamental yet challenging task that groups image pixels with similar features into distinct clusters. Among various approaches, contrastive learning methods, which employ the concept of encouraging semantically similar samples to move closer together while pushing semantically inconsistent samples apart, have garnered significant attention due to their promising performance. However, the most prevalent approaches face two major limitations: 1) treating all samples indiscriminately during optimization, where the abundance of well-categorized samples overwhelms the feature learning process and 2) tending to introduce noise when constructing positive sample pairs through view augmentation or searching the nearest neighbors, which would cause semantic drift of sample features. To solve these issues, we propose a graph autoencoder-based deep clustering framework named spatial–spectral graph contrastive clustering with hard sample mining (SSGCC) that constructs spatial–spectral dual views without data augmentation and focuses more on hard samples rather than treating all samples equally with the aid of spatial–spectral features. Concretely, we extract the spectral features and the neighborhood spatial features of the samples as dual branches to avoid the noise caused by data augmentation and develop the cluster-oriented consistency learning to facilitate the exchange of knowledge between the two spectral–spatial perspectives. In addition, we propose a hard sample mining-based contrastive learning scheme with the aid of spatial–spectral features. To better measure the importance of the samples, we combine spatial features and spectral features to calculate the similarity between sample pairs. The weights of hard sample pairs are dynamically up-weight while the easy ones are down-weighting to improve the discriminative capability. Extensive experiments on four benchmark HSI datasets demonstrate the effectiveness and superiority of the proposed methods against state-of-the-art ones.**

Received 23 July 2024; revised 4 September 2024; accepted 15 September 2024. Date of publication 20 September 2024; date of current version 21 October 2024. This work was supported in part by the National Key Research Development Program of China under Grant 2022ZD0209103 and in part by the National Natural Science Foundation of China under Project 62325604 and Project 62276271. *(Corresponding author: Xinwang Liu.)*

Renxiang Guan, Hao Yu, Dayu Hu, and Xinwang Liu are with the College of Computer, National University of Defense Technology, Changsha 410073, China (e-mail: renxiangguan@nudt.edu.cn; csyuhao@gmail.com; hzauhdy@gmail.com; xinwangliu@nudt.edu.cn).

Wenxuan Tu is with the School of Computer Science and Technology, Hainan University, Haikou 570228, China (e-mail: wenxuantu@163.com).

Zihao Li and Chang Tang are with the School of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: lizihao@cug.edu.cn; tangchang@cug.edu.cn).

Yuzeng Chen and Qiangqiang Yuan are with the School of Geodesy and Geomatics, Wuhan University, Wuhan 430079, China (e-mail: yuzeng_chen@whu.edu.cn; qqyuan@sgg.whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3464648

***Index Terms*— Contrastive learning, deep clustering, graph autoencoder, hard sample mining, hyperspectral images.**

NOMENCLATURE

Notation	Description
$\mathcal{S}$	Set of superpixels.
M	Number of superpixels.
w	Patch size of anchors.
K	Number of clusters.
k	Number of neighbors.
D^S	Dimension of X^S.
$\hat{d}^S$	Dimension of Z^S.
D^P	Dimension of X^P.
$\hat{d}^P$	Dimension of Z^P.
$N = W \times H$	Number of data points.
$\mathbf{X} \in \mathbb{R}^{W \times H \times D}$	HSI cube.
$\mathbf{X}^S \in \mathbb{R}^{M \times D^S}$	Spectral data matrix.
$\mathbf{X}^P \in \mathbb{R}^{M \times D^P}$	Spatial data matrix.
$\mathbf{A}^S \in \mathbb{R}^{M \times M}$	Adjacency matrix of spectral data.
$\mathbf{A}^P \in \mathbb{R}^{M \times M}$	Adjacency matrix of spatial data.
$\mathbf{Z}^S \in \mathbb{R}^{M \times d^S}$	Spectral feature matrix.
$\mathbf{Z}^P \in \mathbb{R}^{M \times d^P}$	Spatial feature matrix.
$\boldsymbol{\mu} \in \mathbb{R}^{K \times \hat{d}}$	Cluster center matrix.
$\mathbf{P} \in \mathbb{R}^M$	Clustering pseudolabels.
$\mathbf{Q} \in \mathbb{R}^{M \times M}$	Cluster pseudo label matrix.
$\mathbf{Y} \in \mathbb{R}^{\bar{M}}$	Set of reliable samples.
$\mathcal{H}$	Similarity function.
$\mathcal{M}$	Weight adjustment function.
$\odot$	Hadamard product.
$\mathcal{F}$	Graph encoder.
$\mathcal{D}$	Graph decoder.
$\Theta_{\mathcal{F}}$	Parameters of graph encoder.
$\Theta_{\mathcal{D}}$	Parameters of graph decoder.
$\|\cdot\|_2$	L-2 norm.
$\|\cdot\|_F$	Frobenius norm.

I. INTRODUCTION

HSI captures detailed spectral information of ground objects that is highly discriminative, along with rich spatial information regarding their locations [1], [2], [3], [4]. Consequently, HSI holds great potential for various

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

applications, including mineral exploration [5], [6], precision agriculture [7], [8], [9], and urban planning [10], [11]. However, due to the high-dimensional HSI data, the interpretation of HSI commonly relies on a significant quantity of high-quality labeled samples to avoid the problem of underfitting resulting from inadequate classifier training [12]. Unfortunately, the manual labeling of samples is commonly time-consuming, which severely restricts the applications of HSI. Therefore, the development of unsupervised clustering techniques becomes essential.

HSI clustering aims to learn discriminative features for grouping each pixel into distinct clusters without relying on labels [13], [14], [15], [16], [17], [18]. Traditional HSI clustering methods primarily aim to optimize the objective function to derive the optimal solution [19], [20], [21], [22]. However, these methods follow a linear assumption and fall short in extracting deep features, which are not applicable to complex HSI scenes [23]. Recently, deep graph clustering methods have gained increasing attention due to the powerful representation capabilities of graph neural networks (GNNs) [24], [25], [26], [27], [28], [29]. Among sets of self-supervised graph clustering techniques [30], [31], [32], contrastive learning, which generates supervision signals to guide the model self-training by enforcing positive samples to be closer together and pushing negative samples apart, has recently made significant breakthroughs in HSI clustering and produced promising results [33], [34], [35], [36], [37], [38]. According to the types of data augmentation employed, existing contrastive learning-based HSI clustering methods can be broadly categorized into two categories. This first category, known as augmentation-dependent methods, typically involves preprocessing techniques such as rotation, cropping, and adding noise to generate augmented samples [33], [34]. However, these strategies may disrupt the original HSI structure and alter the semantic content of the data, potentially leading to a drift in the semantic feature of the samples. In another line of research, the second category, knowns as augmentation-free methods, typically selects the nearest k neighbors of each sample as positive pairs, while the remaining samples are considered as negatives [35], [36]. Nevertheless, the positive samples selected by the augmentation-free method may not always be reliable, which can significantly impact the learning trajectory of the model. Therefore, it is crucial to develop a reliable method for constructing augmentation views that can address the limitations of both augmentation-dependent and augmentation-free strategies.

Moreover, existing contrastive clustering approaches in HSI often face a significant limitation. Despite the potential semantic drift caused by data augmentation, most contrastive learning-based HSI clustering methods assume an equal contribution of all samples to the learning objective. However, this equal treatment optimization strategy can lead to the network training being predominantly influenced by easy samples, thereby neglecting hard yet important samples. In addition, it is susceptible to noise introduced by unlabeled background information. Although there has been research on hard sample mining in general image processing, these methods are not directly applicable to HSI, as they require careful consideration of both spectral information and spatial homogeneity. For instance, in Fig. 1, different classes of samples (e.g., pair 3) may share similar spectral information, while samples from the same class (e.g., pair 4) can exhibit significant spectral differences due to seasonal variations. Furthermore, background pixels in HSI significantly impact unsupervised tasks since they are challenging to distinguish and introduce noise into the learning process, making the clustering of other samples more complex. However, as shown in Fig. 2, most contrastive learning-based HSI clustering methods treat all samples equally. This equal treatment optimization strategy would cause the network pretraining to be strongly guided by easy samples and overlook hard yet important boundary samples [39].

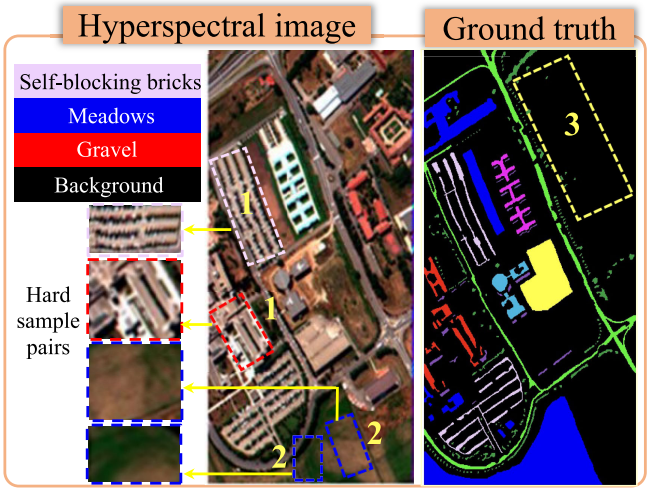


Fig. 1. Definition of hard samples in hyperspectral images can be categorized into three groups: 1) hard positive sample pairs, which belong to the same cluster but exhibit low similarity, such as pair 1; 2) hard negative sample pairs, which belong to different clusters but display high similarity, such as pair 2; and 3) noise samples, referring to background samples that are difficult to distinguish and can potentially interfere with other samples, for instance, sample 3.

To address the aforementioned limitations, we propose a novel method called spatial–spectral graph contrastive clustering with hard sample mining (SSGCC). In this method, we aim to explore hard samples by leveraging spatial–spectral information and enable the model to prioritize these hard samples instead of treating all samples equally. Specifically, we select the central pixels of the superpixels as the anchors and extract the spatial and spectral features of these anchors as dual branches instead of relying on data augmentation. These dual branches are then utilized to learn deep features using graph autoencoders. In this way, the model is enabled to incorporate valuable spatial information to construct dual views while avoiding the introduction of noise issues associated with data augmentation. Subsequently, to facilitate effective knowledge sharing between the compact embeddings obtained from the spatial and spectral perspectives, we introduce clustering-oriented consistency learning to encourage the model to produce consistent similarity distributions for each node, thereby enhancing the semantic richness of the learned node representations. In addition, a novel contrastive learning method based on hard sample mining is proposed to improve

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

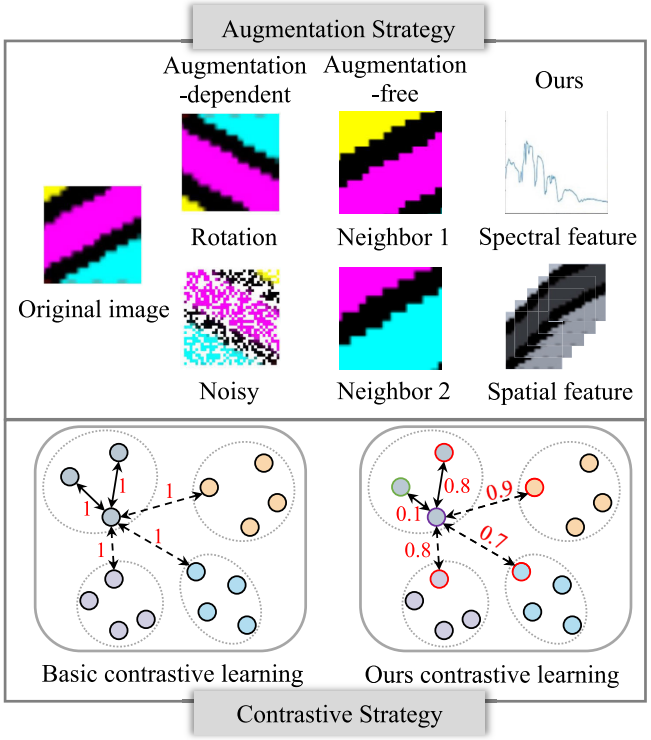


Fig. 2. Our approach differs from existing methods in two main aspects: 1) augmentation strategy: in the method that requires enhancement, augmentation views are constructed using conventional techniques such as adding noise. In contrast, the method that does not require enhancement utilizes the nearest neighbors of each sample as positive samples. In our approach, we extract both spatial and spectral features of the samples to construct a dual view, taking into account both spatial and spectral perspectives and 2) contrastive strategy: previous methods treat all samples equally during the contrastive learning process. In contrast, our method dynamically adjusts the weights between samples, allowing for more flexible and adaptive learning. This adjustment of weights enables us to focus on the importance of different samples and address the issue of disproportionate influence from easy samples encountered in equal treatment optimization strategies.

the discriminative capability of the model. First, we identify samples that exhibit consistent clustering results in both spatial and spectral domains as high-confidence samples. Based on these high-confidence samples and the feature similarity matrix, we design a weight-modulating function that dynamically adjusts the weights between samples. This adaptation allows the model to focus more on positive sample pairs with small similarities and negative sample pairs with large similarities, thus enhancing its attention toward challenging entities. In summary, our work makes the following main contributions.

1) We propose a novel HSI clustering framework termed SSGCC, which makes the first attempt to mine hard sample pairs for HSI.
2) We propose a method for constructing a spatial–spectral view based on superpixel segmentation, and design a novel consistency loss to promote clustering consistency between spatial and spectral perspectives.
3) We propose a hard sample mining-based contrastive learning approach that leverages spatial–spectral features to assist in extracting effective representations and clustering for hard samples.
4) Extensive experiments across four widely used HSI datasets have demonstrated the effectiveness and superiority of the proposed method.

II. Related Works

A. Deep Clustering for Large-Scale HSI

HSI clustering aims to learn discriminative features for grouping each pixel into distinct clusters without relying on labels [1], [40], [41], [42]. Depending on the sizes of the HSI datasets, the current methodologies can be divided into small-scale [38], [43] and large-scale algorithms [26], [35], [44]. Small-scale datasets, characterized by their limited sample sizes, typically represent the specific subsets of scenarios drawn from larger datasets. On the other hand, large-scale datasets pose greater challenges and more accurately reflect real-world data scenarios. Deep learning methods require huge computing power and cannot be well applied to large-scale HSI. To this end, superpixel-based methods, which capitalize on the homogeneity of HSI, are gaining traction [45], [46]. They involve segmenting HSI images into irregularly shaped superpixels before processing. Cai et al. [35] proposed a superpixel pooling autoencoder to efficiently learn contracted subspace representations at the superpixel level. Huang et al. [47] used hierarchical superpixels to combine low-level and high-level semantic information of HSI. In order to aggregate remote spatial information, GCN-based technology has been widely used by researchers in HSI clustering. Zhang et al. [26] introduced a dual graph autoencoder (DGAE) that simultaneously utilizes a superpixel-based similarity graph and a band-based similarity graph. Ding et al. [48] introduced superpixels and a self-correlated learning smoothy enhanced GCN for HSI clustering. However, a common shortfall of such methods is the simplistic averaging of spectral data within superpixels, which fails to fully harness the spatial details of the superpixels.

B. Contrastive Learning in HSI Processing

Recently, contrastive learning has gained popularity in unsupervised learning, which can get rid of the resource intensive annotations [49], [50], [51], [52], [53], [54]. The foundational principle of contrastive learning involves distinguishing between positive and negative samples within a representation space to learn the robust feature. This is done to refine the representation of a given sample, aiming to closely align it with its positive correlate while distinctly segregating it from its negative counterpart. The classical InfoNCE loss [55] is formulated as follows:

$$l = -\log \frac{\exp\left(\text{sim}\left(z_i, z_j\right)\right)}{\exp\left(\text{sim}\left(z_i, z_j\right)\right) + \sum_{k \neq j} \exp(\text{sim}(z_i, z_k))} \quad (1)$$

where $\text{sim}(\cdot)$ denotes the cosine similarity. z_j and z_k form the positive sample pairs and negative sample pairs with z_i, respectively.

Contrastive learning has surfaced as a key technique in the realm of HSI clustering tasks [56], [57], [58], [59], drawing significant research interest. Hu et al. [34] performed contrastive learning via pixel patches and their

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

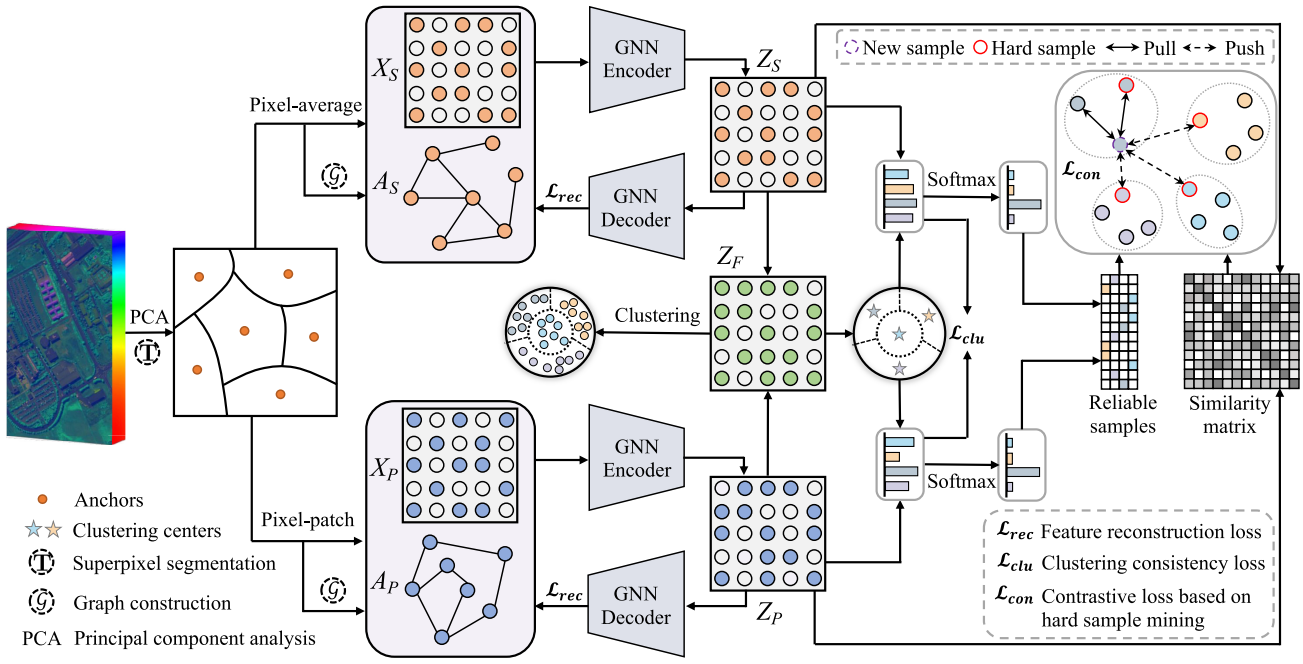


Fig. 3. Illustration of the proposed framework SSGCC. Initially, we select anchors based on the superpixel segmentation results and extract the deep spatial and spectral features of the anchors by graph autoencoders. Then, we design cluster-oriented consistency learning to encourage the model to produce consistent similarity distributions for each node from the two spatial and spectral perspectives. Finally, we propose a hard sample mining-based contrastive learning algorithm with the assistance of spatial–spectral branches. The weights of hard sample pairs are dynamically up-weight while the easy ones are down-weighting to improve the discriminative capability.

noise-based enhanced views. CMSCGC [38] constructed multiviews by extracting view information such as texture of the HSI to promote the proximity of positive samples between different views. Qi et al. [60] ventured into superpixel-level contrastive clustering through the introduction of two data augmentations—pixel sampling and model weight. Cai et al. [35] presented a neighborhood contrastive subspace clustering (NCSC) network, which diverged from view enhancement for positive sample construction and instead identified the nearest samples as positives, treating the rest as negative samples. In addition, Guan et al. [36] combined pixel-superpixel contrastive learning with a pseudolabel correction approach. Despite these developments, a common limitation persists: current methods assume an equal contribution of all samples to the learning objective, affecting their discriminative strength. To bridge this gap, we propose a hard sample mining-based contrastive learning approach that leverages spatial–spectral features to assist in extracting effective representations and clustering for hard samples.

C. Hard Sample Mining in Contrastive Learning

The purpose of hard sample mining in contrastive learning is to screen out samples that are difficult to distinguish and treat them specially, which has been effectively verified in both the image and graph fields [61], [62], [63], [64]. Concretely, Kalantidis et al. [61] proposed to focus on these hard samples in contrastive learning. Zhang et al. [65] proposed to control two hardness-aware properties independently with dual temperature for improving performance. There are also some advanced works based on hard sample mining in the graph field. ProGCL [66] introduced an appropriate measure for the hardness of negative samples by combining similarity with a probability estimator. Tu et al. [39] designed a hierarchically contrastive method, which introduces mutual information technology to help the network find hard samples. Liu et al. [62] designed a hard sample aware network by the assist of both attribute and structure feature. However, these methods are not applicable to HSI. The classification of HSI heavily relies on the spectral features of pixels, which are often complicated by seasonal changes or building reflections that interfere with many types of spectral features, making them hard to distinguish. In addition, HSI contain unique background information that can easily interfere with the extraction of useful information. Therefore, we propose a hard sample mining strategy that combines spatial and spectral information to allow the model to focus more on hard sample pairs, thus enhancing its attention toward challenging entities.

III. Methods

In this section, we introduce the details of the proposed SSGCC model for HSI clustering. The overall framework is illustrated in Fig. 3. Our framework consists of three main components: 1) anchor construction based on superpixel segmentation and spatial–spectral feature extraction based on graph encoder; 2) clustering-oriented consistency learning for semantic enhancing; and 3) contrastive learning algorithm based on hard sample mining. In the following, we will elaborate on each component of our SSGCC in detail.

A. Notations and Definitions

1) Notations: Given an HSI cube $\mathbf{X} \in \mathbb{R}^{W\times H\times D}$, which consists of $N = W \times H$ pixels and D spectral bands, let

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

$\mathcal{G} = (\mathbf{X}, \mathbf{A})$ be the corresponding arbitrary undirected graph. $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the adjacency matrix which is generated based on the similarity of samples. If there is an edge between node v_i and v_j, then $a_{ij} = 1$; otherwise, $a_{ij} = 0$. The normalized adjacency matrix is denoted by $\tilde{\mathbf{A}} = \hat{\mathbf{D}}^{-1/2}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-1/2}$, where $\hat{\hat{A}}$ equal to $\mathbf{A} + \mathbf{I}$, $\mathbf{I}$ is the identity matrix, and $\hat{\mathbf{D}}$ is the diagonal degree matrix. All frequently used notations are listed in the Nomenclature.

2) GCN: Distinct from convolutional neural networks, GCN capitalizes on the neighboring node information to capture distant spatial relationships and glean discriminative features through the aggregation of neighborhood data. The operational essence of a GCN can be articulated as follows:

$$\mathbf{Z}^{(l+1)} = \sigma(\mathbf{S}\mathbf{Z}^{(l)}\mathbf{W}^{(l)}) \tag{2}$$

where $\mathbf{Z}^{(0)}$ is the initial data, $\mathbf{Z}^{(l)}$ is the learned feature, and l is the layer of GCN. $\mathbf{W}^{(l)}$ is the trainable weight matrix of GCN and $\sigma(\cdot)$ is an activation function.

3) HSI Clustering: HSI clustering aims to learn a mapping function $f : \mathbf{X} \mapsto \mathbf{Z} \in \mathbb{R}^{N\times d}$ for grouping each pixel into K distinct clusters without relying on labels, where d is the number of dimension of the latent features. In an optimally clustered outcome, the expectation is that pixels belonging to the same class should be as clustered together as possible, ensuring each cluster remains compact and distinctly segregated from other clusters.

B. Spatial and Spectral Graph Construction

HSI entails a substantial data load that necessitates significant computational resources. From the perspective of spectral dimension, since HSI contains a large amount of redundant spectral information, we use traditional principal component analysis [67] to reduce the dimensionality of the bands. Given the homogeneity of HSI [35], superpixel segmentation techniques can be utilized to decrease the computational demands. In this study, we use the efficient simple linear iterative clustering (SLIC) segmentation [68] method to generate superpixels with adaptive shape and size denoted as $\mathcal{S} = \bigcup_{i=1}^{M} \mathbf{x}_i^{\text{sp}}$, where $M \ll N$ is the number of superpixels and $\mathbf{x}_i^{\text{sp}}$ is the ith superpixels. We select the center pixels of the superpixels as the anchors due to their stronger homogeneity with surrounding pixels. This choice ensures the maximization of complete information inclusion from the superpixels when capturing spatial information subsequently. Different from the previous data augmentation methods, we construct dual views by extracting spatial and spectral features of those anchors, avoiding the noise introduced in the data augmentation process.

1) Spectral Feature: Due to the pixels within a superpixel corresponding to the same category and exhibiting similar spectral features, we employ the spectral average of each pixel to articulate the spectral feature of the anchor. Assume that $\mathbf{X}^S \in \mathbb{R}^{M\times D^S}$ represent the spectral feature of anchors, they can be determined by

$$\mathbf{x}_i^S = \frac{\sum_{j=1}^{n_i} \mathbf{x}_j^i}{n_i}, \quad \mathbf{x}_j^i \in \mathcal{S}_i \tag{3}$$

where n_i is the number of pixels in the ith superpixel and $\mathbf{x}_i^S \in \mathbb{R}^{1\times D^S}$ represent the spectral feature of each anchor.

2) Spatial Feature: To incorporate spatial information, we represent each data point by extracting 3-D patches. Specifically, this entails representing every data point by its central pixel in spatial coordinates along with the surrounding $w \times w$ neighboring pixels. After flatten, the spatial features of anchors can be represented as $\mathbf{X}^P \in \mathbb{R}^{M\times D^P}$.

3) Spatial–Spectral Graph Construction: GCN requires an adjacency matrix as input, so we capture the nearest samples of each anchor as neighbors and construct edges for them, with element $\mathbf{A}_{ij}$ defined as follows:

$$\mathbf{A}_{ij} = \begin{cases} 1, & \text{if } \mathbf{x}_j \in \Gamma(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $\Gamma(\cdot)$ represents the top k nearest samples set of sample. By inputting spatial features and spectral features, respectively, we can obtain two adjacency matrices $\mathbf{A}^S$ and $\mathbf{A}^P \in \mathbb{R}^{M\times M}$.

C. Graph Autoencoder Module

To fully explore the topology of the constructed graph and aggregate features from neighboring nodes, we deploy two graph encoders $\mathbf{Z}^S$ and $\mathbf{Z}^P$ to capture features. Concretely, we utilize a multilayer GCN as a graph encoder and update nodes embedding by transporting the original feature on each graph. The procedure could be defined as follows:

$$\mathbf{Z}^S = \mathcal{F}_S\left(\widetilde{\mathbf{A}}^S, \mathbf{X}^S \mid \Theta_{\mathcal{F}_S}\right) \tag{5}$$

$$\mathbf{Z}^P = \mathcal{F}_P\left(\widetilde{\mathbf{A}}^P, \mathbf{X}^P \mid \Theta_{\mathcal{F}_P}\right) \tag{6}$$

where $\Theta_{\mathcal{F}_S}$ and $\Theta_{\mathcal{F}_P}$ denote the learnable parameters of two graph encoders. $\widetilde{\mathbf{A}} \in \mathbb{R}^{M\times M}$ denotes the normalized adjacency matrix.

Within the graph encoder module, building upon the enhancements detailed in [17], we employ a multilayer GCN as for the decoder $D(\cdot)$ to reconstruct the original spatial and spectral information. The process of reconstruction features is articulated as follows:

$$\hat{\mathbf{X}}^S = \mathcal{D}_S\left(\widetilde{\mathbf{A}}^S, \mathbf{Z}^S \mid \Theta_{\mathcal{D}_S}\right) \tag{7}$$

$$\hat{\mathbf{X}}^P = \mathcal{D}_P\left(\widetilde{\mathbf{A}}^P, \mathbf{Z}^P \mid \Theta_{\mathcal{D}_P}\right) \tag{8}$$

where $\hat{\mathbf{X}}^S$ and $\hat{\mathbf{X}}^P$ are the reconstruction features, $\Theta_{\mathcal{D}_S}$ and $\Theta_{\mathcal{D}_P}$ denote the learnable parameters of two graph encoders. Therefore, the individual-level reconstruction loss function could be defined as follows:

$$\mathcal{L}_{\text{rec}} = \left\|\mathbf{X}^S - \hat{\mathbf{X}}^S\right\|_F^2 + \left\|\mathbf{X}^P - \hat{\mathbf{X}}^P\right\|_F^2 \tag{9}$$

where $\|\cdot\|_F$ represents the Frobenius norm.

D. Cluster-Oriented Consistency Learning

We develop the cluster-oriented consistency learning to facilitate the exchange of knowledge between the two spatial–spectral perspectives. Meanwhile, we expect that the final features can finely reflect the inherent relationships among samples, thereby enhancing the label-free HSI clustering. Toward this end, we share the compact semantics by comparing the similarities of each node to other samples in the embedding spaces of the two views.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

In detail, we first use the attention-based weight fusion method to obtain the fused latent feature Z^F

$$\mathbf{Z}^F = \mathbf{W}_1 \odot \mathbf{Z}^S + \mathbf{W}_2 \odot \mathbf{Z}^P \tag{10}$$

where $\odot$ is the Hadamard product and $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{M\times\hat{d}}$ are the trainable weight matrices to control the importance of spatial–spectral features.

More precisely, we compute the fused features $\mathbf{Z}^F$ of the dual branches and employ the K-means algorithm $\mathcal{K}(\cdot)$ to ascertain the clustering centroid of the fused feature

$$\{\boldsymbol{\mu}, \mathbf{P}\} \leftarrow \mathcal{K}(\mathbf{Z}^F) \tag{11}$$

where $\boldsymbol{\mu} \in \mathbb{R}^{K\times\hat{d}}$ denotes the cluster center matrix and $\mathbf{P} \in \mathbb{R}^M$ is the clustering pseudo labels. Then, we first measure the similarity between the ith node embedding $\mathbf{z}_i$ and the kth cluster centroid $\mathbf{c}_k$ by student's t-distribution as follows:

$$q_{ij}^S = \frac{\left(1 + \left\|\mathbf{z}_i^S - \boldsymbol{\mu}_j\right\|^2/\eta\right)^{\left(-\frac{\eta+1}{2}\right)}}{\sum_j^K \left(1 + \left\|\mathbf{z}_i^S - \boldsymbol{\mu}_j\right\|^2/\eta\right)^{\left(-\frac{\eta+1}{2}\right)}} \tag{12}$$

$$q_{ij}^P = \frac{\left(1 + \left\|\mathbf{z}_i^P - \boldsymbol{\mu}_j\right\|^2/\eta\right)^{\left(-\frac{\eta+1}{2}\right)}}{\sum_j^K \left(1 + \left\|\mathbf{z}_i^P - \boldsymbol{\mu}_j\right\|^2/\eta\right)^{\left(-\frac{\eta+1}{2}\right)}} \tag{13}$$

where q_{ij} refers to the probability of assigning ith node to the jth cluster. Here, η denotes the degrees of freedom, and we set $\eta = 1$ in all of our experiments.

With the spectral and spatial view similarity distributions, we encourage their consistency to facilitate knowledge transfer and mutually enhance the representation semantics. Formally, we define the consistency learning loss as follows:

$$\mathcal{L}_{\text{clu}} = \frac{1}{2M}\sum_{i=1}^{M}\sum_{j=1}^{K}\left(\text{KL}\left(\mathbf{q}_{ij}^S \| \mathbf{q}_{ij}^P\right) + \text{KL}\left(\mathbf{q}_{ij}^P \| \mathbf{q}_{ij}^S\right)\right) \tag{14}$$

where $\text{KL}(\cdot\|\cdot)$ is the Kullback–Leibler (KL) divergence. In the training, we minimize $\mathcal{L}_{\text{clu}}$ to optimize our proposed SSGCC and enhance self-supervised learning.

Guided by the cluster consistency constraint outlined in (14), we identify and record those samples for which the labels predicted by the spectral and spatial branches concur as reliable samples set $\mathbf{Y} \in \mathbb{R}^{\bar{M}}$, where $\bar{M}<M$ is the number of reliable samples. Based on clustering pseudolabels $\mathbf{P}$, we calculate the sample pair pseudolabels $\mathbf{Q} \in \mathbb{R}^{M\times M}$ as follows:

$$\mathbf{Q}_{ij} = \begin{cases} 1, & \mathbf{P}_i = \mathbf{P}_j \\ 0, & \mathbf{P}_i \neq \mathbf{P}_j. \end{cases} \tag{15}$$

Here, a value of $\mathbf{Q}_{ij} = 1$ indicates that ith and jth samples belong to the same cluster, suggesting they are more likely to constitute a positive sample pair. Conversely, they are most likely to represent a negative sample pair.

Algorithm 1 Learning Procedure for SSGCC

Input: HSI data $\mathbf{X}$, the input size w, the number of neighbors k, the number of superpixels M, ant the training epochs I
Output: Clustering results

```
Initialize the parameters of the network.
Contruct superpixel set S by using superpixel segmentation T.
Construct the spectral and spatial features of the anchors based on the pixel feature average and sliding window.
Construct spectral matrix A^S and spatial adjacency matrix A^P by Eq. (4).
for i = 1 to ≤ I do
    Obtain Z^S and Z^P with f_S and f_P by Eq. (5) and Eq. (6).
    Obtain fused feature Z^F by Eq. (10).
    Obtain pseudo labels Q by Eq. (15).
    Design similarity function H by Eq. (16).
    Design weight modulating function M by Eq. (17).
    Calculate reconstruction loss L_rec by Eq. (9).
    Calculate clustering consistency loss L_clu by Eq. (14).
    Calculate hard sample mining-based contrastive loss L_con by Eq. (18).
    Update parameters of the network by miniziming Eq. (19).
end for
Using K-means on fused feature Z^F to get the cluster result P.
return P
```

E. Hard Sample Mining-Based Contrastive Learning

The classic contrastive loss depicted in (1) treats all samples uniformly, neglecting the exploration of challenging samples and consequently leading to less than optimal outcomes. In this section, we innovate a hard sample mining contrastive learning network that leverages reliable samples $\mathbf{Y}$, cluster pseudolabels $\mathbf{Q}$, and the similarity function $\mathcal{H}$ to mine hard samples. Also, the $\mathcal{H}$ is computed by

$$\mathcal{H}(i,j) = \frac{\left(\mathbf{Z}_i^S\right)^\top \mathbf{Z}_j^F}{2\cdot\|\mathbf{Z}_i^S\|\cdot\|\mathbf{Z}_j^F\|} + \frac{\left(\mathbf{Z}_i^P\right)^\top \mathbf{Z}_j^F}{2\cdot\|\mathbf{Z}_i^P\|\cdot\|\mathbf{Z}_j^F\|} \tag{16}$$

where $i, j \in \{1, 2, \ldots, M\}$. Leveraging reliable samples $\mathbf{Y}$, cluster pseudolabels $\mathbf{Q}$, and the similarity function $\mathcal{H}$, we have devised a weight adjustment function $\mathcal{M}$ to dynamically modulate the weights of sample pairs throughout the training process. The calculation of $\mathcal{M}$ is as follows:

$$\mathcal{M}(i,j) = \begin{cases} 1, & i \notin \mathbf{Y} \cup j \notin \mathbf{Y} \\ |\mathbf{Q}_{ij} - \text{Norm}(\mathcal{H}(i,j))|^\beta, & \text{others} \end{cases} \tag{17}$$

where Norm denotes the min-max normalization and $\beta \geq 1$ is a scaling factor. When not all samples exhibit high confidence ($i \notin \mathbf{Y} \cup j \notin \mathbf{Y}$), we resort to using the classic infoNCE loss function. However, in cases where all samples are of high confidence, our method is adept at dynamically adjusting weights based on the similarity and pseudoinformation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.



Fig. 4. False color images of the datasets and visualizations of superpixel segmentation. (a) False-color image of Salinas, RGB: [32, 21, 11]. (b) Segmentation of Salinas. (c) False-color image of Pavia University, RGB: [55, 41, 12]. (d) Segmentation of Pavia University. (e) False-color image of Trento, RGB: [60, 24, 3]. (f) Segmentation of Trento. (g) False-color image of XuZhou, RGB: [30, 13, 1]. (h) Segmentation of XuZhou.

We analyze the crucial role of the function $\mathcal{M}$ in hard sample mining. As illustrated in Fig. 1, hard samples are defined as: 1) samples within the same cluster but with low similarity (hard positive samples) and 2) samples in different clusters but with high similarity (hard negative samples). The function $\mathcal{M}$ assigns higher weights to sample pairs that meet these conditions, while simpler sample pairs receive lower weights. Consequently, $\mathcal{M}$ up-weights positive sample pairs with low similarity (the hard ones) and down-weights those with high similarity (the easy ones).

Based on $\mathcal{H}$ and $\mathcal{M}$, we formulate the hard sample mining-based contrastive loss as follows:

$$\mathcal{L}_{\text{con}} = \frac{1}{M}\sum_{i,j=1}^{M} -\log\left(\frac{e^{\mathcal{M}(i,i)\cdot\mathcal{H}(i,i)}}{e^{\mathcal{M}(i,i)\cdot\mathcal{H}(i,i)}+\sum_{i\neq j} e^{\mathcal{M}(i,j)\cdot\mathcal{H}(i,j)}}\right). \tag{18}$$

Compared with the previous contrastive learning-based methods that treat all samples equally, we designed the hard sample mining method which makes the network to focus more on hard sample pairs with the assist of spectral–spatial information. Also, the weights of hard sample pairs are dynamically up-weight while the easy ones are down-weighting to improve the discriminative capability.

F. Overall Loss Function

In summary, we introduce a novel SSGCC for large HSI. During the training stage, the graph encoders, cluster space consistency module, and hard sample mining-based contrastive learning regularization are jointly optimized according to the objective function as follows:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \sigma_1 * \mathcal{L}_{\text{clu}} + \sigma_2 * \mathcal{L}_{\text{con}} \tag{19}$$

where σ_1 and σ_2 are trade-off parameters. The detailed learning procedure of the proposed SSGCC is described in Algorithm 1.

G. Complexity Analysis

We analyze the complexity analysis from the two aspects of model structure and loss computation. To simplify calculations, we define the original feature dimensions as D, the dimensions of each layer of graph autoencoder as $d_1, \ldots, d_L$, and define latent feature dimensions as $\hat{d}$. The complexities of graph encoder is $\mathcal{O}(|E|\sum_{i=1}^{L} d_i d_{i-1})$, where $|E|$ is the number of edge. The time complexities of clustering consistency loss are $\mathcal{O}(MK\hat{d})$, where K is the number of class. Finally, the time complexity of constructing $\mathcal{H}$ and $\mathcal{M}$ are $\mathcal{O}(M^2\hat{d})$. Thus, the overall complexity is $\mathcal{O}(|E|\sum_{i=1}^{L} d_i d_{i-1} + MK\hat{d} + M^2\hat{d})$. Since $K \ll M$, the overall computational complexity is approximately $\mathcal{O}(|E|\sum_{i=1}^{L} d_i d_{i-1} + M^2\hat{d})$.

IV. Experiments

A. Experimental Setup

1) Datasets: To facilitate a comprehensive comparison of the models proposed in this article, we select four foundational datasets. Also, the false color images of these datasets and visualizations of superpixel segmentation are shown in Fig. 4.

1) *Pavia University:* It is gathered from an aerial operation over Pavia, Northern Italy, in 2002, utilizing the Reflectance Optical System Imaging Spectrometer (ROSIS). After excluding 12 bands identified as noisy, the dataset features 103 spectral bands. It showcases a spatial array of 610 × 340 pixels, coupled with a spatial resolution of 1.3 m, illustrating a comprehensive urban landscape. After removing the background, the Pavia University dataset has a total of 19 classes and 42 766 valid labels.
2) *Salinas:* It was collected in 1998 and initially comprised 224 spectral bands spanning the 360–2500-nm wavelength spectrum. The dataset presents imagery with spatial dimensions of 512 × 217 pixels. After the exclusion of bands with significant absorption, the dataset is refined to 204 spectral bands. After removing the background, the Salina dataset has a total of 16 classes and 54 129 valid labels.
3) *Trento:* The Trento dataset originates from the rural surroundings of Trento, Italy. It encompasses a total of 600 × 166 pixels, comprising 63 spectral bands that span wavelengths from 420.89 to 989.09 nm. After removing the background, the Trento dataset has a total of six classes and 30 214 valid labels.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

TABLE I
HYPERPARAMETER SETTINGS

Parameters		Salinas	Pavia University	Trento	Xu
Setting	M	2000	3000	1750	1000
	w	11	11	9	7
	σ_1	10	1000	1	1
	σ_2	10	1000	1	1
	β	3	3	2	2
	k	45	50	50	45
Range	M	{500, ..., 3000}, {1500, ..., 4000}, {500, ..., 1750}			
	w	{1, 3, 5, 7, 9, 11}			
	σ_1	{0.1, 1, 10, 100, 1000, 10000}			
	σ_2	{0.1, 1, 10, 100, 1000, 10000}			
	β	{1, 2, 3, 4}			
	k	{15, 20, 25, 30, 35, 40, 45, 50, 55, 60}			

4) *XuZhou:* The XuZhou dataset is the product of an aerial survey conducted using the HYSPEX hyperspectral camera over XuZhou's fringe urban areas. The dataset has 436 spectral bands and encapsulates a variety of 500 × 260 pixel remote sensing images, capturing a transitional urban landscape. After removing the background, the XuZhou dataset has a total of nine classes and 68 877 valid labels.

2) Compared Methods: To show the effectiveness of our proposed SSGCC, we compare it with 16 clustering methods as follows.

1) *K-means (TPAMI'02)* [41] is an outstanding representative of traditional unsupervised learning.
2) *FCM (SSBM'13)* [40] is a fuzzy clustering method based on an objective function.
3) *SSC (TAPMI'13)* [69] is a classic sparse subspace clustering algorithm.
4) *AEC (ICML'12)* [70] is a deep clustering algorithm based on autoencoder.
5) *DEC (ICML'16)* [71] learns deep features and cluster results using deep encoders.
6) *IDEC (IDEC'17)* [72] improves deep embedded clustering with local structure preservation.
7) *S^5C (NeurIPS'19)* [21] is a selective sampling-based scalable sparse subspace clustering.
8) *HESSC (RS'20)* [42] proposes a hierarchical clustering algorithm that analyzes HSI in a robust and fast manner.
9) *MS^2A-Net (JSTARS'22)* [15] proposes a multiscale spectral–spatial association network to cluster HSI.
10) *DGAE (TCSVT'22)* [26] achieves HSI clustering by designing a pixel-level graph autoencoder and a band-level graph autoencoder.
11) *NCSC (TGRS'22)* [35] introduces a neighborhood contrastive subspace clustering network for HSI clustering.
12) *AGCSC (CVPR'23)* [20] proposes an adaptive graph convolutional subspace clustering algorithm.
13) *DSCRLE (TGRS'23)* [14] designs a deep spectral clustering with regularized linear embedding for HSI clustering.
14) *PSCPC (ICASSP'24)* [36] uses a pixel–superpixel contrastive learning with a pseudolabel correction approach.
15) *CMSCGC (TGRS'24)* [38] introduces multiview information and contrastive learning for HSI clustering.
16) *SDST (TGRS'24)* [44] designs a self-supervised double-structure transformer for HSI clustering.

3) Implement Details: To ensure an equitable comparison between the proposed SSGCC and these baselines, we compute the average results from ten iterative runs under identical experimental conditions, utilizing a 24-GB RTX 3090 GPU and 64 GB of RAM. Key parameters of the SSGCC are detailed in Table I. The training epoch number is set to 100, and we conduct ten runs for all methods. We set the learning rate and weight decay to 1×10^{-3} and 5×10^{-4}, respectively. In additional, we employ the Adam optimizer for model optimization. We use a three-layer GCN as the encoder and decoder of the graph, with the number of hidden layers and output layers being 128, 256, and 512, respectively. For the baselines, we use the optimal parameters as reported in the papers, alongside the corresponding open-source code, to derive the final outcomes. In the case of AGCSC and CMSCGC methods, given their limitations with large-scale datasets, we have introduced a data preprocessing component, specifically employing superpixels as their input.

To clearly demonstrate the performance, we employ four evaluation metrics: overall accuracy (ACC), Kappa, normalized mutual information (NMI), and adjusted rand index (ARI) [73], [74], [75], [76], [77], [78], [79], [80]. Notably, ACC and NMI range in [0, 1] while Kappa and ARI have value ranges of $[-1, 1]$.

B. Experimental Results

We compare our proposed SSGCC competitive HSI clustering baselines and the comparison clustering results are shown in Table II. From the results, we can have the following key observations.

1) The proposed SSGCC significantly outperforms all the compared methods, achieving the highest metrics values on four benchmarks. For example, the impressive achievement strongly proves the effectiveness of the proposed spatial–spectral graph construction and the hard sample mining-based contrastive learning model to mine hard sample pairs and adaptively adjust their weights.
2) Compared with classical graph clustering methods, contrastive clustering approaches have shown superior performance. The reason lies in the ability of contrastive learning to more effectively exploit intrinsic semantic information. By learning discriminative representations in a principled manner, contrastive learning better serves the clustering task. Furthermore, our method's ability to identify hard samples leads to a performance enhancement over other contrastive learning methods. Compared with NCSC and CMSCGC, the proposed SSGCC produces ACC performance gains of 7.02%/9.83%, 12.67%/5.81%, 24.77%/ 6.85%, and 6.71%/8.68% across four datasets.
3) GCN-based methods outperform traditional approaches by capturing richer information through the aggregation of neighboring features. Within the realm of machine learning algorithms, AGCSC significantly outshines conventional methods like K-means and S^5C

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLUSTERING RESULTS ON FOUR HSI DATASETS. THE BEST AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED WITH BOLD AND UNDERLINE, RESPECTIVELY

No.	Salinas				Pavia University				Trento				XuZhou			
	ACC	Kappa	NMI	ARI	ACC	Kappa	NMI	ARI	ACC	Kappa	NMI	ARI	ACC	Kappa	NMI	ARI
K-means	61.11	57.58	71.24	48.20	51.74	41.99	52.14	31.17	62.81	52.96	46.18	34.26	40.42	29.58	42.40	26.60
FCM	55.98	52.26	68.77	43.65	42.73	33.79	49.95	31.17	50.68	41.27	46.18	28.00	39.91	31.32	42.24	24.87
SSC	65.46	62.34	76.47	58.02	52.04	41.57	42.13	33.46	76.59	69.65	66.92	71.70	69.74	63.23	57.58	57.88
AEC	63.54	59.32	76.77	57.62	45.75	35.25	51.76	32.41	77.56	70.85	77.20	79.84	66.66	59.09	64.97	65.96
DEC	57.22	51.46	72.78	54.52	45.14	35.22	54.01	28.82	64.88	53.66	66.00	58.50	49.65	37.34	42.40	36.66
IDEC	62.21	57.91	75.50	55.45	44.79	32.95	48.48	32.34	72.99	66.03	78.57	74.93	67.75	60.35	65.84	67.90
S^{5}C	53.51	49.34	62.08	39.52	48.36	37.73	43.23	25.08	47.42	32.19	38.68	23.70	63.14	55.04	52.26	51.57
HESSC	52.93	47.87	60.03	41.50	47.71	36.26	49.43	29.27	49.94	30.31	32.83	21.92	42.35	26.69	32.42	25.49
MS^{2}A-Net	64.63	60.31	74.47	59.95	60.40	48.15	53.36	44.77	83.60	78.14	76.24	81.28	62.61	53.77	58.21	55.28
DGAE	59.50	55.60	70.89	47.87	52.65	42.89	47.55	31.93	79.85	73.00	65.88	70.65	67.61	59.93	61.05	61.36
NCSC	74.97	72.07	79.69	66.51	51.07	42.11	45.31	32.81	66.85	58.23	63.07	58.06	74.03	68.33	72.39	67.95
AGCSC	63.84	59.78	76.65	55.91	57.63	46.99	47.60	36.57	65.52	56.63	59.13	53.91	76.07	70.39	69.52	67.77
DSCRLE	57.13	52.89	60.86	35.94	48.50	36.54	44.92	22.68	77.11	69.04	65.69	70.31	71.06	64.96	62.28	56.74
PSCPC	65.19	60.21	61.08	65.53	59.36	45.16	50.44	58.41	78.26	82.44	74.35	74.70	74.14	67.86	68.48	73.03
CMSCGC	72.16	66.29	69.16	77.93	57.93	43.46	49.19	57.57	84.77	81.90	79.88	84.81	72.06	66.62	66.01	72.36
SDST	74.33	71.30	80.11	61.86	50.70	39.17	42.66	29.19	82.96	76.55	71.92	74.90	67.00	57.72	62.82	60.15
Ours	**81.99**	**76.36**	**80.30**	**86.21**	**63.74**	**52.86**	**54.88**	**59.27**	**91.62**	**93.57**	**88.77**	**87.80**	**80.74**	**74.92**	**76.29**	**77.93**

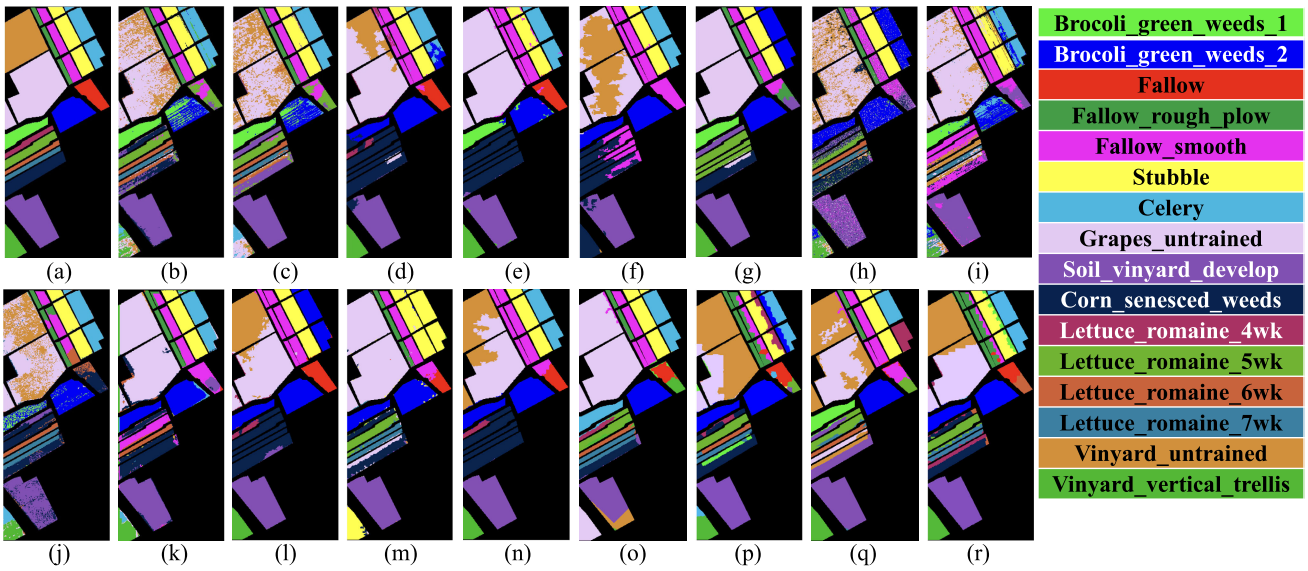


Fig. 5. Clustering maps on Salinas dataset. (a) Ground truth, (b) K-means, (c) FCM, (d) SSC, (e) AE, (f) DEC, (g) IDEC, (h) S^{5}C, (i) HESSC, (j) MS^{2}A-Net, (k) DGAE, (l) NCSC, (m) AGCSC, (n) DSCRLE, (o) PSCPC, (p) CMSCGC, (q) SDST, and (r) our proposed SSGCC.

by adaptively sourcing neighbor information via graph convolutional operations. In the domain of deep learning algorithms, our method exhibits substantial improvements over CNN-based (PSCPC) and transform-based methods (SDST). Specifically, compared to PSCPC and SDST, SSGCC secures ACC performance boosts of 16.8%/7.66%, 4.38%/13.04%, 13.36%/8.66%, and 6.6%/13.74%, showing its enhanced effectiveness.

C. Qualitative Comparison

Figs. 5–8 display the clustering images generated by the proposed SSGCC model and baseline models across four datasets. It is evident that our model delivers superior visualization outcomes, with the ability to distinctly differentiate most types and present clearer boundaries. Notably, in the Salinas dataset, while other models struggle with misclassifying the four types of lettuce romaine, our model distinguishes them with clarity. Similarly, results depicted in Fig. 7 for the Trento dataset illustrate our model's proficiency in accurately clustering vineyard and wood. In comparison, other algorithms not only tend to confuse these sample points but also fail to delineate clear boundaries at their intersections. The exceptional performance of SSGCC is largely attributed to its capacity to identify and concentrate on challenging samples, thereby significantly reducing the likelihood of misclassification.

We also plot the distributions of the learned embeddings reduced by t-distributed stochastic neighbor embedding (t-SNE) in Fig. 9 to visually verify the validity of our proposed SSGCC. We can see that SSGCC exhibits better separability among different clusters, with improved aggregation and diversity within the same cluster and a larger gap between different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

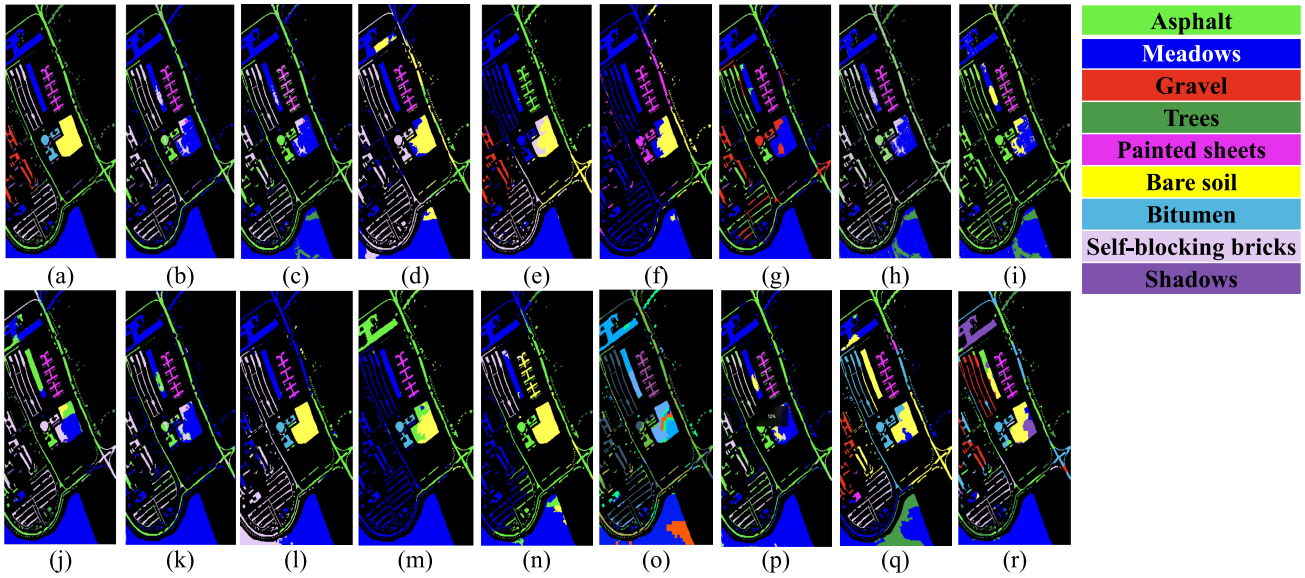


Fig. 6. Clustering maps on Pavia University dataset. (a) Ground truth, (b) K-means, (c) FCM, (d) SSC, (e) AE, (f) DEC, (g) IDEC, (h) S^5C, (i) HESSC, (j) MS^2A-Net, (k) DGAE, (l) NCSC, (m) AGCSC, (n) DSCRLE, (o) PSCPC, (p) CMSCGC, (q) SDST, and (r) our proposed SSGCC.

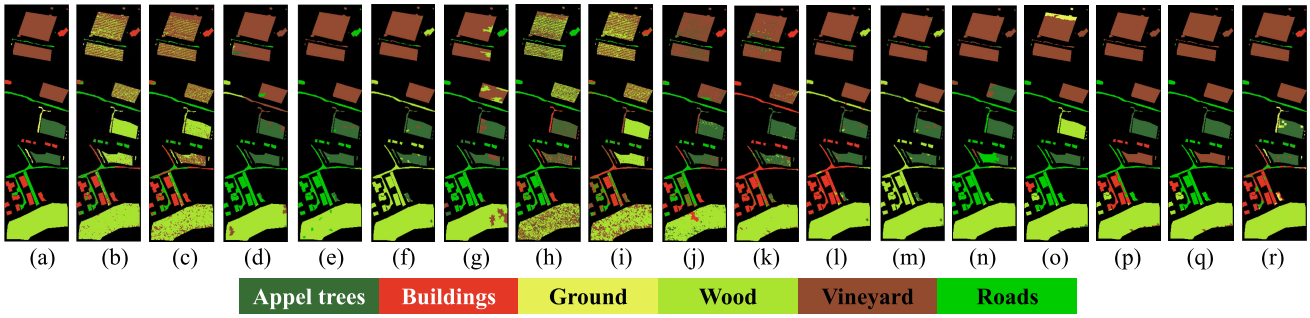


Fig. 7. Clustering maps on Trento dataset. (a) Ground truth, (b) K-means, (c) FCM, (d) SSC, (e) AE, (f) DEC, (g) IDEC, (h) S^5C, (i) HESSC, (j) MS^2A-Net, (k) DGAE, (l) NCSC, (m) AGCSC, (n) DSCRLE, (o) PSCPC, (p) CMSCGC, (q) SDST, and (r) our proposed SSGCC.

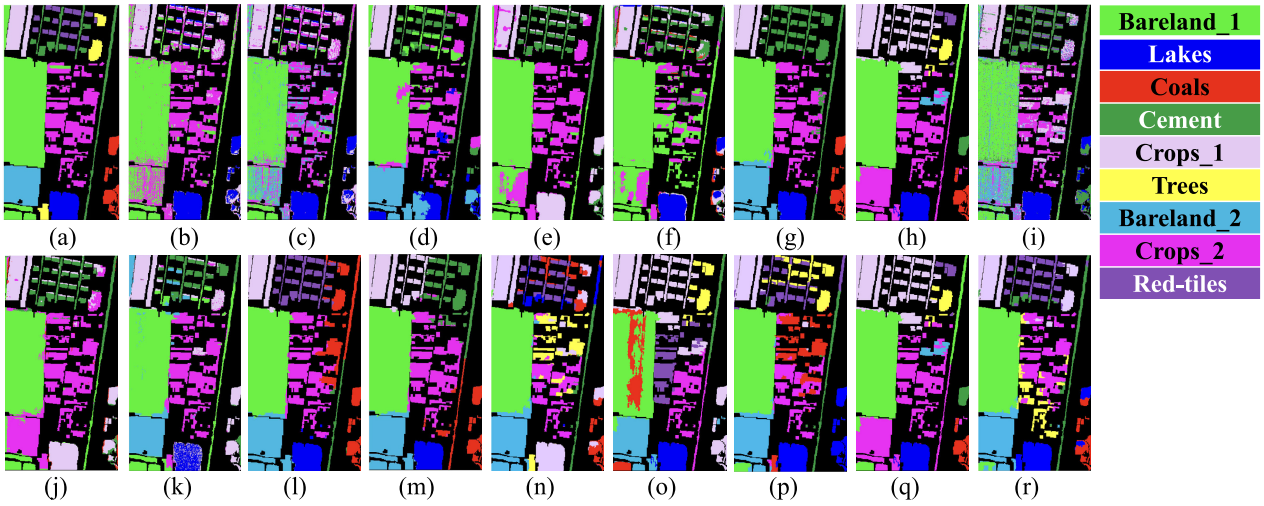


Fig. 8. Clustering maps on XuZhou dataset. (a) Ground truth, (b) K-means, (c) FCM, (d) SSC, (e) AE, (f) DEC, (g) IDEC, (h) S^5C, (i) HESSC, (j) MS^2A-Net, (k) DGAE, (l) NCSC, (m) AGCSC, (n) DSCRLE, (o) PSCPC, (p) CMSCGC, (q) SDST, and (r) our proposed SSGCC.

clusters, showcasing its ability to learn more discriminative representations.

D. Running Time

To illustrate the efficiency of the proposed SSGCC, we compared its runtime with that of a deep learning-based HSI clustering algorithm across four datasets. Table III demonstrates that SSGCC exhibits significant efficiency advantages. Methods such as AE, DEC, and IDEC are slower because they process pixel-level data and require batch processing to incrementally input data. In contrast, while NCSC and CMSCGC use superpixels for data preprocessing, subspace clustering involves additional iterative steps and more complex optimization processes compared with K-means.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RUNNING TIME (SECOND) OF DEEP LEARNING-BASED METHODS ON FOUR DATASETS

Dataset	AE	DEC	IDEC	HESSC	MS^{2}A-Net	DGAE	NCSC	DSCRLE	PSCPC	CMSCGC	SDST	Ours
Salinas	949	794	586	730	488	46	500	125	92	622	986	58
Pavia University	824	647	373	1014	152	43	621	235	190	735	925	116
Trento	524	443	190	555	106	39	435	124	86	831	552	36
XuZhou	1113	950	754	960	473	117	420	538	101	573	1842	39

TABLE IV
ABLATION EXPERIMENTS ON SSGCC IN THREE ASPECTS: MAIN LOSS, AUGMENTATION STRATEGY, AND CONTRASTIVE SCHEME

Dataset	Metric	Main Loss			Augmentation Strategy			Contrastive Scheme	
		w/o $\mathcal{L}_{\text{rec}}$	w/o $\mathcal{L}_{\text{clu}}$	w/o $\mathcal{L}_{\text{con}}$	Feature mask	Feature noise	Edge permutation	w/o $\mathcal{M}$	Ours
Salinas	ACC	71.28±3.64	80.45±0.65	73.75±2.55	75.83±2.92	76.87±2.76	76.29±2.34	78.39±0.98	**81.99±0.84**
	NMI	68.17±4.07	78.65±0.70	71.41±2.68	73.53±3.08	74.61±2.98	74.17±2.48	76.30±1.06	**80.30±0.93**
	ARI	77.36±1.75	86.83±0.56	80.61±1.18	81.52±1.27	82.08±1.27	82.98±0.96	84.83±0.81	**86.21±1.08**
	Kappa	64.87±6.82	76.76±0.89	71.42±4.50	72.55±4.85	75.45±5.04	73.44±2.78	73.57±2.22	**76.36±3.20**
PaviaU	ACC	51.15±7.48	62.52±1.33	54.02±6.94	56.28±3.56	58.55±3.82	59.40±2.63	59.66±4.66	**63.74±1.36**
	NMI	42.28±7.10	52.34±2.31	44.77±6.76	47.42±3.56	49.29±3.87	49.70±2.92	50.69±4.83	**54.88±1.72**
	ARI	49.42±4.21	56.67±2.05	52.25±3.09	52.68±3.32	54.64±2.89	54.84±2.25	56.45±5.18	**59.27±1.78**
	Kappa	39.0±10.95	52.07±5.32	44.49±9.52	45.91±7.63	45.71±7.59	47.78±4.64	46.26±9.59	**52.86±1.93**
Trento	ACC	77.36±0.25	88.77±0.93	81.30±0.37	84.76±0.85	86.90±0.82	84.44±4.27	87.85±1.46	**91.62±0.88**
	NMI	73.30±0.32	85.04±1.16	78.37±0.50	81.08±1.12	83.25±1.10	79.72±5.29	83.86±1.91	**88.77±1.15**
	ARI	72.78±0.45	84.12±0.86	76.60±0.80	80.16±0.96	82.83±0.72	81.71±2.02	82.77±1.34	**87.80±0.97**
	Kappa	77.39±0.49	90.72±2.07	82.75±0.65	84.61±1.03	88.66±1.75	84.82±5.73	88.66±1.86	**93.57±1.13**
XuZhou	ACC	74.14±1.15	77.92±1.08	75.58±0.56	75.55±1.00	76.70±0.77	77.91±0.74	78.23±1.12	**80.74±0.89**
	NMI	68.48±1.37	72.94±1.34	70.21±0.64	70.81±1.27	72.08±0.90	73.13±0.86	73.34±1.37	**76.29±1.04**
	ARI	73.03±1.34	74.85±0.67	75.94±0.45	73.12±1.36	74.44±1.04	76.02±0.80	75.24±0.48	**77.93±1.07**
	Kappa	67.86±1.09	69.53±0.38	69.34±1.12	69.78±2.18	70.11±2.21	71.83±2.13	69.69±0.38	**74.92±2.33**

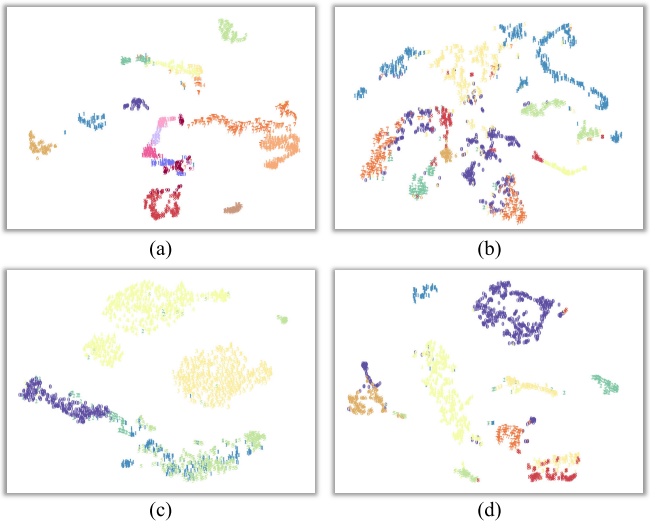


Fig. 9. t-SNE visualization of the proposed SSGCC on the four datasets. (a) Salinas. (b) Pavia University. (c) Trento. (d) XuZhou.

These subspace clustering-based algorithms typically have an algorithmic complexity of $\mathcal{O}(N^3)$, which, despite enhancing ACC, also results in increased time consumption. Similarly, although SDST can reduce the spatial correlation in large datasets, its high complexity leads to substantial time overhead. In summary, the proposed SSGCC demonstrates superior runtime efficiency due to its optimized approach, which effectively balances ACC and computational complexity.

E. Ablation Study

We conduct ablation experiments from three aspects: main loss, augmentation strategy, and contrastive learning. By selectively removing specific elements or changing different augmentation strategies, we aim to evaluate their individual contributions and understand their influence on overall performance. The results are summarized in Table IV.

1) Ablation on the Main Loss: We define w/o $\mathcal{L}_{\text{rec}}$, w/o $\mathcal{L}_{\text{clu}}$, and w/o $\mathcal{L}_{\text{con}}$ as the proposed SSGCC variants without reconstruction loss, clustering consistent loss and hard sample mining-based contrastive learning loss, respectively. The results indicated that the MCGC method exhibited the best performance. When removing the clustering consistency loss and contrast loss, the ACC of the model drops by 1.54%/8.24%, 1.22%/9.72, 2.85%/10.32%, and 2.82%/5.16% on the four datasets, respectively. This indicates that each module of the network is effective and their absence could result in performance degradation.

2) Ablation on the Augmentation strategy: In order to avoid the noise effect caused by complex augmentation strategies, we extract the spatial–spectral features of the samples to construct the dual view. In this section, we compare SSGCC with some classic data augmentation methods such as edge dropping and edge dropping. Specifically, we adopt the data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

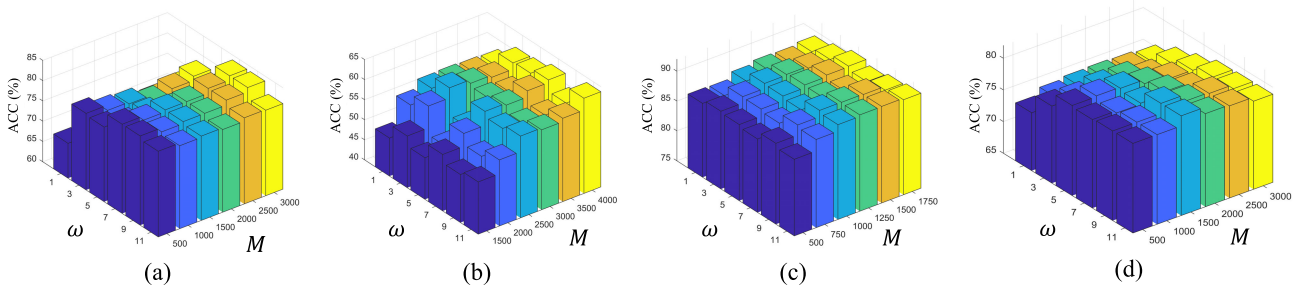


Fig. 10. Sensitivity analysis of two hyperparameters M and w. (a) Salinas. (b) Pavia University. (c) Trento. (d) XuZhou.

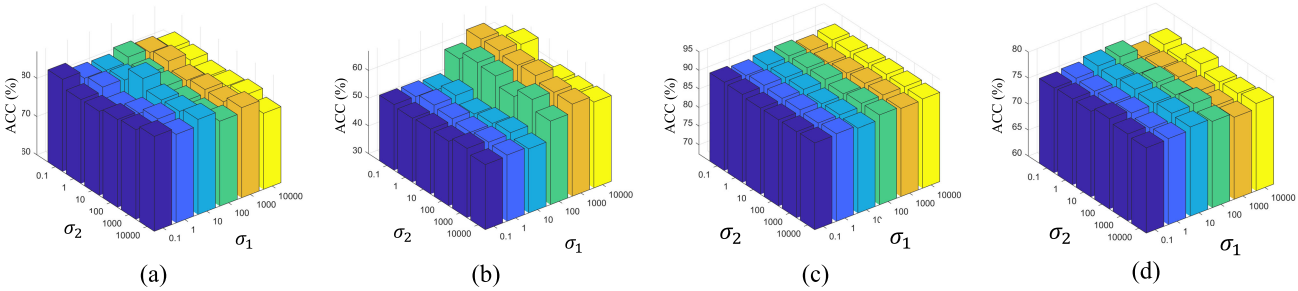


Fig. 11. Sensitivity analysis of two hyperparameters σ_1 and σ_2. (a) Salinas. (b) Pavia University. (c) Trento. (d) XuZhou.

augmentation as randomly masking 30% features ("feature mask") or randomly adding 30% edges ("feature noise"), or randomly perturbing 30% edges ("edge permutation"). From the results, we can see that these methods have an average decrease in ACC of 2%–6%, indicating that these classic data augmentation strategies will introduce noise and cause semantic drift. These results also show the effectiveness of our method of extracting spatial–spectral features for contrastive learning.

3) Ablation on the Contrastive Scheme: To demonstrate the effectiveness of our proposed contrastive learning method based on hard sample mining, we compare the basic contrastive learning (w/o $\mathcal{M}$) with our proposed SSGCC. From the results, we can observe that compared with the baseline, our contrastive learning strategy has increased by 3.6%, 4.07%, 3.32%, and 2.51%, respectively. This shows that our dynamic weight strategy can effectively mine hard samples and improve the model's discriminative ability.

F. Parameter Analysis

1) Impact of Hyperparameters w and M: The hyperparameter w controls the neighborhood size of the anchors, and the hyperparameter M is used to control the number of superpixels. We choose the center point of the superpixel as the anchor, so the number of superpixels is equal to the number of anchor. Our observations from Fig. 10 are as follows. First, on the Pavia and XuZhou datasets, the number of superpixels has a greater impact on the model ACC. This may be because the homogeneity of these two datasets is relatively small and requires more refined segmentation. Second, the introduced neighborhood spatial information can greatly improve the ACC of the model. When there is only spectral information (w =1), the ACC on the four datasets is poor. As the neighborhood increases, more spatial information is captured, providing reliable spatial view information for the model. However, when w is too large, the model ACC will also decrease. This may be because too large a neighborhood value will remove the noise information contained in the other analog sample points.

2) Impact of Hyperparameter σ: Equation (19) introduces hyperparameters σ_1 and σ_2 to balance the importance of clustering consistency loss and contrastive learning loss, respectively. To show its influence in-depth, Fig. 11 presents the ACC variation of SSGCC on four datasets when σ varies from 0.1 to 10 000. It can be observed that our method achieves relatively stable performance within a certain range of σ_1 and σ_2, indicating that our SSGCC is insensitive to these parameters on some datasets. In the Pavia University dataset, a notable improvement in model ACC is observed when the value of $\sigma_2 \geq 100$. This enhancement likely stems from the low homogeneity of Pavia University dataset and its composition of more challenging samples. The hard sample mining strategy we have proposed significantly contributes to this improvement, markedly enhancing the clustering performance.

3) Impact of Hyperparameter β: The hyperparameter β controls weight rate of sample pairs. Fig. 12 shows the four evaluation metric of SSGCC on four datasets when β varies from 1 to 4 with a step size of 1. When β is too small, simple samples still have a large weight and the model performance is low. However, when β is too small, the weight of simple samples is almost 0, and the model ACC also decreases, indicating that completely ignoring simple samples also has a negative impact on model performance. Finally, we set β to 2, 2, 3, and 3 on the four datasets.

4) Impact of Hyperparameter k: The hyperparameter k is the number of neighbors. Fig. 13 shows the ACC of SSGCC

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

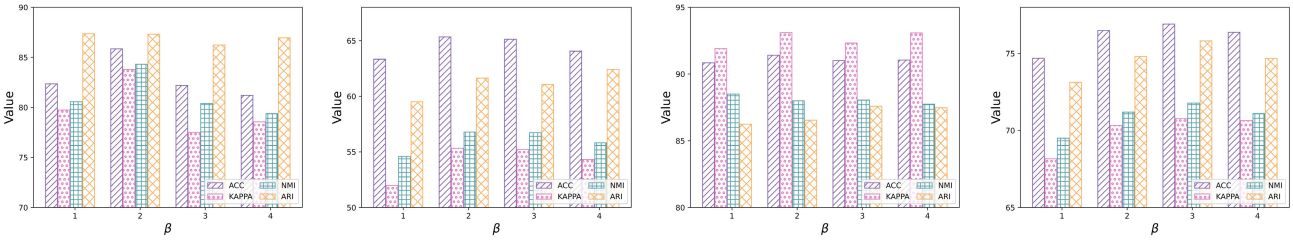


Fig. 12. Sensitivity analysis of hyperparameter β. (a) Salinas. (b) Pavia University. (c) Trento. (d) XuZhou.

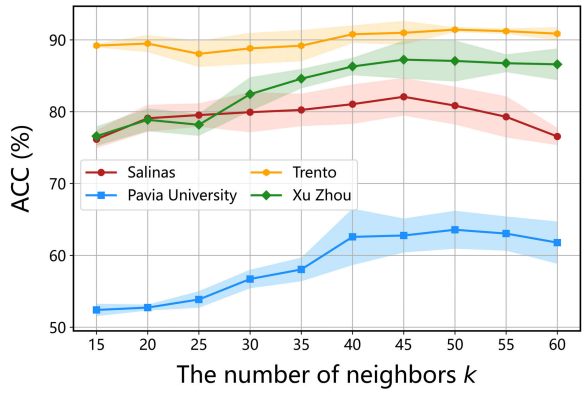


Fig. 13. Sensitivity analysis of hyperparameter k.

on four datasets when k varies from 15 to 60 with a step size of 5. The findings indicate that when k is too small, the model misses crucial neighbor information, leading to an insufficient capacity to capture remote spatial information, which negatively impacts ACC. Conversely, when k is too large, it tends to introduce noise, which also degrades the model's performance. Finally, we set k to 45, 50, 50, and 45 on the four datasets.

V. Conclusion

In this article, we propose a novel approach named SSGCC for deep HSI clustering. Specifically, SSGCC uses superpixel segmentation results to select anchor points and uses GAE to capture spatial–spectral features. Subsequently, based on the spatial–spectral fusion features, SSGCC performs cluster-oriented consistency learning of the node similarities from spatial and spectral views, which enables node representations with richer semantics. Finally, we propose a contrastive learning algorithm based on hard sample mining to dynamically set the weights of sample pairs and enable the model to prioritize these hard samples instead of treating all samples equally. Experiments show that SSGCC can outperform other clustering methods on four HSI datasets. In the future, we will explore remote sensing image clustering algorithms under the coordination of multimodal information, which aims to boost the performance of clustering by utilizing complementary knowledge from multimodal data.

References

[1] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[2] Y. Chen, Q. Yuan, Y. Tang, Y. Xiao, J. He, and Z. Liu, "SENSE: Hyperspectral video object tracker via fusing material and motion cues," *Inf. Fusion*, vol. 109, Sep. 2024, Art. no. 102395.

[3] J. Liu, R. Guan, Z. Li, J. Zhang, Y. Hu, and X. Wang, "Adaptive multi-feature fusion graph convolutional network for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 23, p. 5483, Nov. 2023.

[4] C. Wang, Y. Guo, and J. Fu, "Dual-stream discriminative attention network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515512.

[5] Z. Chen et al., "Global to local: A hierarchical detection algorithm for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544915.

[6] X. Zhang, Z. Wang, J. Li, and Z. Hua, "MVAFG: Multiview fusion and advanced feature guidance change detection network for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11050–11068, 2024.

[7] D. Yao et al., "Deep hybrid: Multi-graph neural network collaboration for hyperspectral image classification," *Defence Technol.*, vol. 23, pp. 164–176, May 2023.

[8] R. Guan, Z. Li, T. Li, X. Li, J. Yang, and W. Chen, "Classification of heterogeneous mining areas based on ResCapsNet and GaoFen-5 imagery," *Remote Sens.*, vol. 14, no. 13, p. 3216, Jul. 2022.

[9] Y. Chen, Q. Yuan, Y. Tang, Y. Xiao, J. He, and L. Zhang, "SPIRIT: Spectral awareness interaction network with dynamic template for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503116.

[10] R. Huang, Y. Xu, D. Hong, W. Yao, P. Ghamisi, and U. Stilla, "Deep point embedding for urban classification using ALS point clouds: A new perspective from local to global," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 62–81, May 2020.

[11] Y. Chen et al., "SSTtrack: A unified hyperspectral video tracking framework via modeling spectral–spatial–temporal conditions," *Inf. Fusion*, vol. 114, Feb. 2025, Art. no. 102658.

[12] H. Zhai, H. Zhang, P. Li, and L. Zhang, "Hyperspectral image clustering: Current achievements and future lines," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 35–67, Dec. 2021.

[13] Y. Qin, L. Bruzzone, and B. Li, "Learning discriminative embedding for hyperspectral image clustering based on set-to-set and sample-to-sample distances," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 473–485, Jan. 2020.

[14] Y. Zhao and X. Li, "Deep spectral clustering with regularized linear embedding for hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509311.

[15] K. R. Shahi, P. Ghamisi, B. Rasti, R. Gloaguen, and P. Scheunders, "MS^{2}A-Net: Multiscale spectral–spatial association network for hyperspectral image clustering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6518–6530, 2022.

[16] Z. Hao, Z. Lu, G. Li, F. Nie, R. Wang, and X. Li, "Ensemble clustering with attentional representation," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 2, pp. 581–593, Feb. 2023.

[17] W. Tu et al., "Deep fusion clustering network," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 11, pp. 9978–9987.

[18] F. Luo, Y. Liu, X. Gong, Z. Nan, and T. Guo, "EMVCC: Enhanced multi-view contrastive clustering for hyperspectral images," in *Proc. ACM Multimedia*, 2024.

[19] S. Liu and H. Wang, "Graph convolutional optimal transport for hyperspectral image spectral clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4414013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

[20] L. Wei, Z. Chen, J. Yin, C. Zhu, R. Zhou, and J. Liu, "Adaptive graph convolutional subspace clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Sep. 2023, pp. 6262–6271.

[21] S. Matsushima and M. Brbic, "Selective sampling-based scalable sparse subspace clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.

[22] J. Chen, S. Liu, Z. Zhang, and H. Wang, "Diffusion subspace clustering for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1–15, 2023.

[23] Y. Cai, M. Zeng, Z. Cai, X. Liu, and Z. Zhang, "Graph regularized residual subspace clustering network for hyperspectral image clustering," *Inf. Sci.*, vol. 578, pp. 85–101, Nov. 2021.

[24] Y. Ding et al., "Self-supervised locality preserving low-pass graph convolutional embedding for large-scale hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536016.

[25] T. Han, S. Niu, X. Gao, W. Yu, N. Cui, and J. Dong, "Deep low-rank graph convolutional subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533513.

[26] Y. Zhang, Y. Wang, X. Chen, X. Jiang, and Y. Zhou, "Spectral–spatial feature extraction with dual graph autoencoder for hyperspectral image clustering," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 12, pp. 8500–8511, Dec. 2022.

[27] B. Peng, Y. Yao, J. Lei, L. Fang, and Q. Huang, "Graph-based structural deep spectral–spatial clustering for hyperspectral image," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.

[28] Z. Zhang et al., "Multireceptive field: An adaptive path aggregation graph neural framework for hyperspectral image classification," *Expert Syst. Appl.*, vol. 217, May 2023, Art. no. 119508.

[29] Y. Ding et al., "Multi-scale receptive fields: Graph attention neural network for hyperspectral image classification," *Expert Syst. Appl.*, vol. 223, Aug. 2023, Art. no. 119858.

[30] Y. Cai, Z. Zhang, X. Liu, Y. Ding, F. Li, and J. Tan, "Learning unified anchor graph for joint clustering of hyperspectral and LiDAR data," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–14, 2024, doi: 10.1109/TNNLS.2024.3392484.

[31] W. Tu et al., "Initializing then refining: A simple graph attribute imputation network," in *Proc. 31st Int. Joint Conf. Artif. Intell.*, 2022, pp. 3494–3500.

[32] W. Tu et al., "RARE: Robust masked graph autoencoder," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 10, pp. 5340–5353, Oct. 2023.

[33] Y. Cai et al., "Large-scale hyperspectral image clustering using contrastive learning," 2021, *arXiv:2111.07945*.

[34] X. Hu, T. Li, T. Zhou, and Y. Peng, "Deep spatial–spectral subspace clustering for hyperspectral images based on contrastive learning," *Remote Sens.*, vol. 13, no. 21, p. 4418, 2021.

[35] Y. Cai et al., "Superpixel contracted neighborhood contrastive subspace clustering network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530113.

[36] R. Guan, Z. Li, X. Li, and C. Tang, "Pixel-superpixel contrastive learning and pseudo-label correction for hyperspectral image clustering," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 6795–6799.

[37] S. Huang, H. Zeng, H. Chen, and H. Zhang, "Spatial and cluster structural prior guided subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511115.

[38] R. Guan et al., "Contrastive multiview subspace clustering of hyperspectral images based on graph convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510514.

[39] W. Tu, S. Zhou, X. Liu, C. Ge, Z. Cai, and Y. Liu, "Hierarchically contrastive hard sample mining for graph self-supervised pretraining," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–14, 2023, doi: 10.1109/TNNLS.2023.3297607.

[40] J. C. Bezdek, *Pattern Recognition With Fuzzy Objective Function Algorithms*. Cham, Switzerland: Springer, 2013.

[41] T. Kanungo, D. M. Mount, N. S. Netanyahu, C. D. Piatko, R. Silverman, and A. Y. Wu, "An efficient k-means clustering algorithm: Analysis and implementation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 881–892, Jul. 2002.

[42] K. R. Shahi, M. Khodadadzadeh, L. Tusa, P. Ghamisi, R. Tolosana-Delgado, and R. Gloaguen, "Hierarchical sparse subspace clustering (HESSC): An automatic approach for hyperspectral image analysis," *Remote Sens.*, vol. 12, no. 15, p. 2421, Jul. 2020.

[43] K. Huang, X. Li, Y. Pi, H. Cheng, and G. Xu, "Graph-constrained residual self-expressive subspace clustering network for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 941–955, 2024.

[44] F. Luo, Y. Liu, Y. Duan, T. Guo, L. Zhang, and B. Du, "SDST: Self-supervised double-structure transformer for hyperspectral images clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511614.

[45] K. Cui et al., "Superpixel-based and spatially regularized diffusion learning for unsupervised hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4405818.

[46] H. Zhao, F. Zhou, L. Bruzzone, R. Guan, and C. Yang, "Superpixel-level global and local similarity graph-based clustering for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519316.

[47] N. Huang, L. Xiao, Q. Liu, and J. Chanussot, "S2DMSC: A self-supervised deep multilevel subspace clustering approach for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512817.

[48] Y. Ding et al., "Unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536716.

[49] Z. Zhang et al., "Graph contrastive learning as a versatile foundation for advanced scRNA-seq data analysis," *bioRxiv*, Jan. 2024.

[50] Y. Liu, Z. Zhang, S. Qin, F. D. Salim, and A. J. Yepes, "Contrastive learning-based imputation-prediction networks for in-hospital mortality risk modeling using EHRs," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*, 2023, pp. 428–443.

[51] Y. Liu et al., "Simple contrastive graph clustering," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 13789–13800, Oct. 2023.

[52] W. Tu et al., "Attribute-missing graph clustering network," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 14, pp. 15392–15401.

[53] D. Hu, Z. Dong, K. Liang, H. Yu, S. Wang, and X. Liu, "High-order topology for deep single-cell multiview fuzzy clustering," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 8, pp. 4448–4459, Aug. 2024.

[54] D. Hu, K. Liang, Z. Dong, J. Wang, Y. Zhao, and K. He, "Effective multi-modal clustering method via skip aggregation network for parallel scRNA-seq and scATAC-seq data," *Briefings Bioinf.*, vol. 25, no. 2, Jan. 2024, Art. no. bbae102.

[55] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[56] R. Guan, Z. Li, C. Song, G. Yu, X. Li, and R. Feng, "S2RC-GCN: A spatial–spectral reliable contrastive graph convolutional network for complex land cover classification using hyperspectral images," 2024, *arXiv:2404.00964*.

[57] J. Wang et al., "Multi-level graph subspace contrastive learning for hyperspectral image clustering," 2024, *arXiv:2404.05211*.

[58] J. Li, "Superpixel-based dual-neighborhood contrastive graph autoencoder for deep subspace clustering of hyperspectral image," in *Proc. Int. Conf. Intell. Comput.* Cham, Switzerland: Springer, 2024, pp. 181–192.

[59] Y. Cai, Z. Zhang, P. Ghamisi, B. Rasti, X. Liu, and Z. Cai, "Transformer-based contrastive prototypical clustering for multimodal remote sensing data," *Inf. Sci.*, vol. 649, Nov. 2023, Art. no. 119655.

[60] J. Qi, Y. Jia, H. Liu, and J. Hou, "Superpixel graph contrastive clustering with semantic-invariant augmentations for hyperspectral images," 2024, *arXiv:2403.01799*.

[61] Y. Kalantidis, M. B. Sariyildiz, N. Pion, P. Weinzaepfel, and D. Larlus, "Hard negative mixing for contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21798–21809.

[62] Y. Liu et al., "Hard sample aware network for contrastive deep graph clustering," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, 2023, pp. 8914–8922.

[63] K. Liang et al., "Knowledge graph contrastive learning based on relation-symmetrical structure," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 1, pp. 226–238, Aug. 2023.

[64] Z. Hao, H. Xin, L. Wei, L. Tang, R. Wang, and F. Nie, "Towards expansive and adaptive hard negative mining: Graph contrastive learning via subspace preserving," in *Proc. ACM Web Conf.*, vol. 33, May 2024, pp. 322–333.

[65] C. Zhang et al., "Dual temperature helps contrastive learning without many negative samples: Towards understanding and simplifying MoCo," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14421–14430.

[66] J. Xia, L. Wu, G. Wang, J. Chen, and S. Z. Li, "ProGCL: Rethinking hard negative mining in graph contrastive learning," in *Proc. ICML*, vol. 162, 2022, pp. 24332–24346.

[67] C. Rodarmel and J. Shan, "Principal component analysis for hyperspectral image classification," *Surv. Land Inf. Syst.*, vol. 62, no. 2, pp. 115–122, Jun. 2002.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

[68] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
[69] E. Elhamifar and R. Vidal, "Sparse subspace clustering: Algorithm, theory, and applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2765–2781, Nov. 2013.
[70] P. Baldi, "Autoencoders, unsupervised learning, and deep architectures," in *Proc. ICML Workshop Unsupervised Transf. Learn.*, 2012, pp. 37–49.
[71] J. Xie, R. Girshick, and A. Farhadi, "Unsupervised deep embedding for clustering analysis," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 478–487.
[72] X. Guo, L. Gao, X. Liu, and J. Yin, "Improved deep embedded clustering with local structure preservation," in *Proc. 26th Int. Joint Conf. Artif. Intell.*, Aug. 2017, pp. 1753–1759.
[73] H. Quan, X. Li, D. Hu, T. Nan, and X. Cui, "Dual-channel prototype network for few-shot pathology image classification," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 7, pp. 4132–4144, Jul. 2024.
[74] L. Meng et al., "SARF: Aliasing relation–assisted self-supervised learning for few-shot relation reasoning," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–11, 2024, doi: 10.1109/TNNLS.2024.3355151.
[75] M. Liu et al., "Deep temporal graph clustering," in *Proc. 12th Int. Conf. Learn. Represent.*, 2024.
[76] M. Liu et al., "Self-supervised temporal graph learning with temporal and structural intensity alignment," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–13, 2024, doi: 10.1109/TNNLS.2024.3386168.
[77] K. Liang et al., "A survey of knowledge graph reasoning on graph types: Static, dynamic, and multi-modal," *IEEE Trans. Pattern Anal. Mach. Intell.*, pp. 1–20, 2024, doi: doi: 10.1109/TPAMI.2024.3417451.
[78] R. Li et al., "SMWE-GFPNNet: A high-precision and robust method for forest fire smoke detection," *Knowl.-Based Syst.*, vol. 289, Apr. 2024, Art. no. 111528.
[79] T. Hu, Q. Yan, Y. Qi, and Y. Zhang, "Generating content for HDR deghosting from frequency view," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 36, Jun. 2024, pp. 25732–25741.
[80] Q. Yan et al., "Toward high-quality HDR deghosting with conditional diffusion models," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 4011–4026, May 2024.



Renxiang Guan received the B.S. degree from China University of Geosciences (CUG), Wuhan, China, in 2023. He is currently pursuing the master's degree with the National University of Dense Technology (NUDT), Changsha, China.

He has published several peer-reviewed articles, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), Association for the Advancement of Artificial Intelligence Learned Society (AAAI), International Conference on Acoustics, Speech, and Signal Processing (ICASSP), *Knowledge-Based Systems*, IJCNN, and *Remote Sensing*. His research interests include deep graph clustering and hyperspectral image processing.



Wenxuan Tu received the Ph.D. degree from the School of Computer, National University of Defense (NUDT), Changsha, China, in 2023.

He is currently an Associate Professor with the School of Computer Science and Technology, Hainan University, Haikou, China. He has published several papers in highly regarded journals and conferences, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), NeurIPS, International Conference on Machine Learning (ICML), International Conference on Learning Representations (ICLR), Association for the Advancement of Artificial Intelligence (AAAI), and IJCAI. His research interests include clustering analysis and graph machine learning.



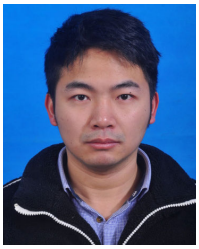
Zihao Li received the B.S. degree from China University of Geosciences (CUG), Wuhan, China, in 2023, where he is currently pursuing the Ph.D. degree.

He has published several peer-reviewed articles, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), International Conference on Acoustics, Speech, and Signal Processing (ICASSP), and IJCNN. His research interests include remote sensing image process, computer vision, and deep learning.



Hao Yu (Student Member, IEEE) received the B.Eng. degree in computer science from Inner Mongolia University, Hohhot, China, in 2019, and, the M.A.Sc. degree in cyberspace science and technology from Beijing Institute of Technology, Beijing, China, in 2022. He is currently pursuing the Ph.D. degree with the National University of Defense Technology (NUDT), Changsha, China.

He has authored several papers in top-level journals and conferences, such as IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY (TIFS), TDSC, and ACM MM, and served as a reviewer for highly regarded journals, such as TIFS, IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), and ACM TOIS. His research focuses on AI security and federated learning.



Dayu Hu received the B.Sc. degree from Northeastern University (NEU), Shenyang, China, in 2022. He is currently pursuing the Ph.D. degree with the National University of Defense Technology (NUDT), Changsha, China.

He has published several papers and served as a PC member/reviewer in highly regarded journals and conferences such as ACM MM, AAAI, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), and TCBB. His research interests include graph learning and bioinformatics.



Yuzeng Chen received the B.S. degree in geographic information science from Southwest University of Science and Technology, Mianyang, China, in 2020, and the M.S. degree from Central South University, Changsha, China, in 2023. He is currently pursuing the Ph.D. degree with the School of Geodesy and Geomatics, Wuhan University, Wuhan, China.

His research interests include remote sensing/hyperspectral video processing and computer vision. More details can be found at: https://yzcu.github.io.



Chang Tang (Senior Member, IEEE) received the Ph.D. degree from Tianjin University, Tianjin, China, in 2016.

He was with the AMRL Laboratory, University of Wollongong, Wollongong, NSW, Australia, from 2014 to 2015. He is currently a Full Professor with the School of Computer Science, China University of Geosciences, Wuhan, China. He has published more than 100 peer-reviewed papers, including those in highly regarded journals and conferences such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON HUMAN-MACHINE SYSTEMS (THMS), ICCV, CVPR, IJCAI, AAAI, and ACM MM. His research interests include machine learning and computer vision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.

Dr. Tang won the Best Paper Award at the Fifth Asian Conference on Artificial Intelligence Technology. He serves as an Associate Editor for *Neural Networks*, *CAAI Transactions on Intelligence Technology*, and *BMC Bioinformatics*. He regularly serves on the technical program committees or as the Area Chair for some top conferences such as NIPS, ICML, CVPR, ICCV, ECCV, IJCAI, ICME, and AAAI.



Qiangqiang Yuan (Member, IEEE) received the B.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2006 and 2012, respectively.

In 2012, he joined the School of Geodesy and Geomatics, Wuhan University, where he is currently a Professor. He has published more than 90 research articles, including more than 70 peer-reviewed articles in international journals, such as IEEE TRANSACTIONS ON IMAGE PROCESSING and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include image reconstruction, remote sensing image processing and application, and data fusion.

Dr. Yuan was a recipient of the Top-Ten Academic Star of Wuhan University in 2011 and the Youth Talent Support Program of China in 2019. In 2014, he received the Hong Kong Scholar Award from the Society of Hong Kong Scholars and China National Postdoctoral Council. He is on the editorial board of nine international journals and has frequently served as a referee for more than 50 international journals for remote sensing and image processing.



Xinwang Liu (Senior Member, IEEE) received the Ph.D. degree from the National University of Defense Technology (NUDT), Changsha, China, in 2013.

He is currently a Professor with the School of Computer, NUDT. He has published more than 100 peer-reviewed papers, including those in highly regarded journals and conferences such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY (TIFS), ICML, NeurIPS, ICCV, CVPR, AAAI, and IJCAI. His research interests include kernel learning and unsupervised feature learning.

Dr. Liu serves as an Associate Editor for TNNLS, IEEE TRANSACTIONS ON CYBERNETICS (TCYB), and *Information Fusion Journal*. More information can be found at: https://xinwangliu.github.io/.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:07:34 UTC from IEEE Xplore. Restrictions apply.